



STATUS OF THE PROTON THERAPY COMPLEX PROMETHEUS

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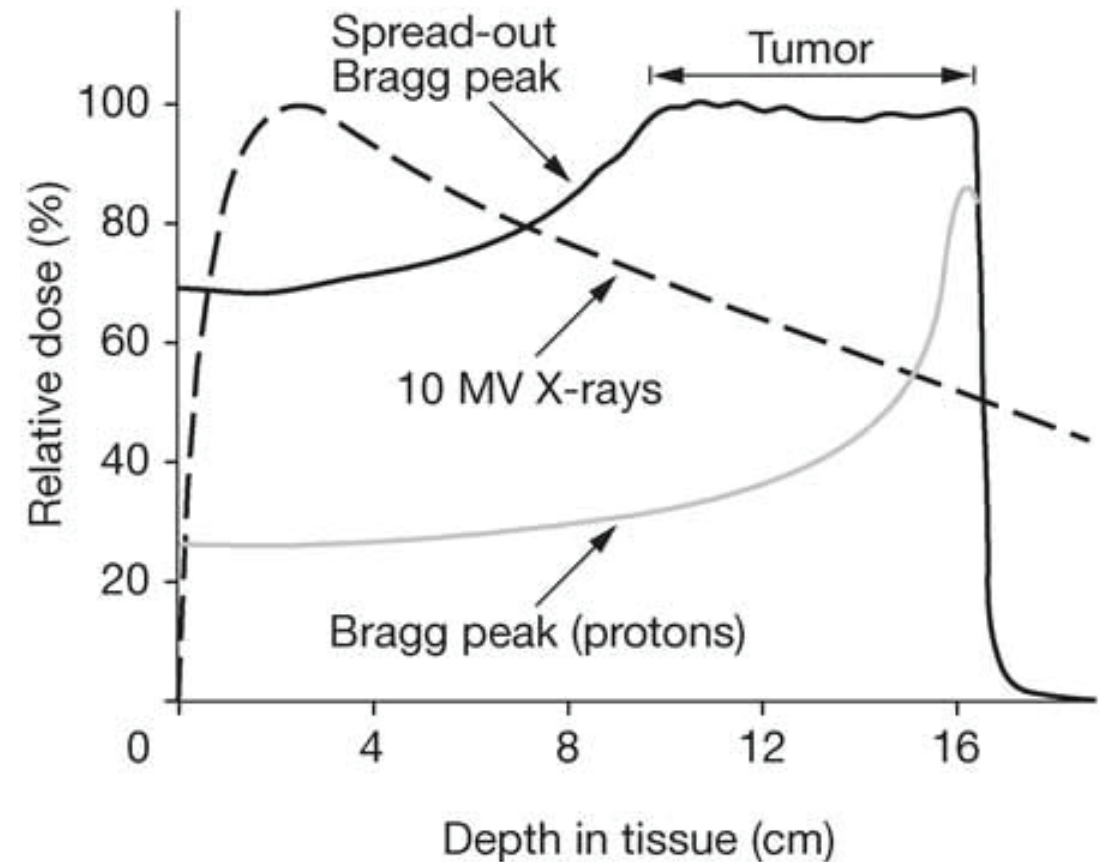
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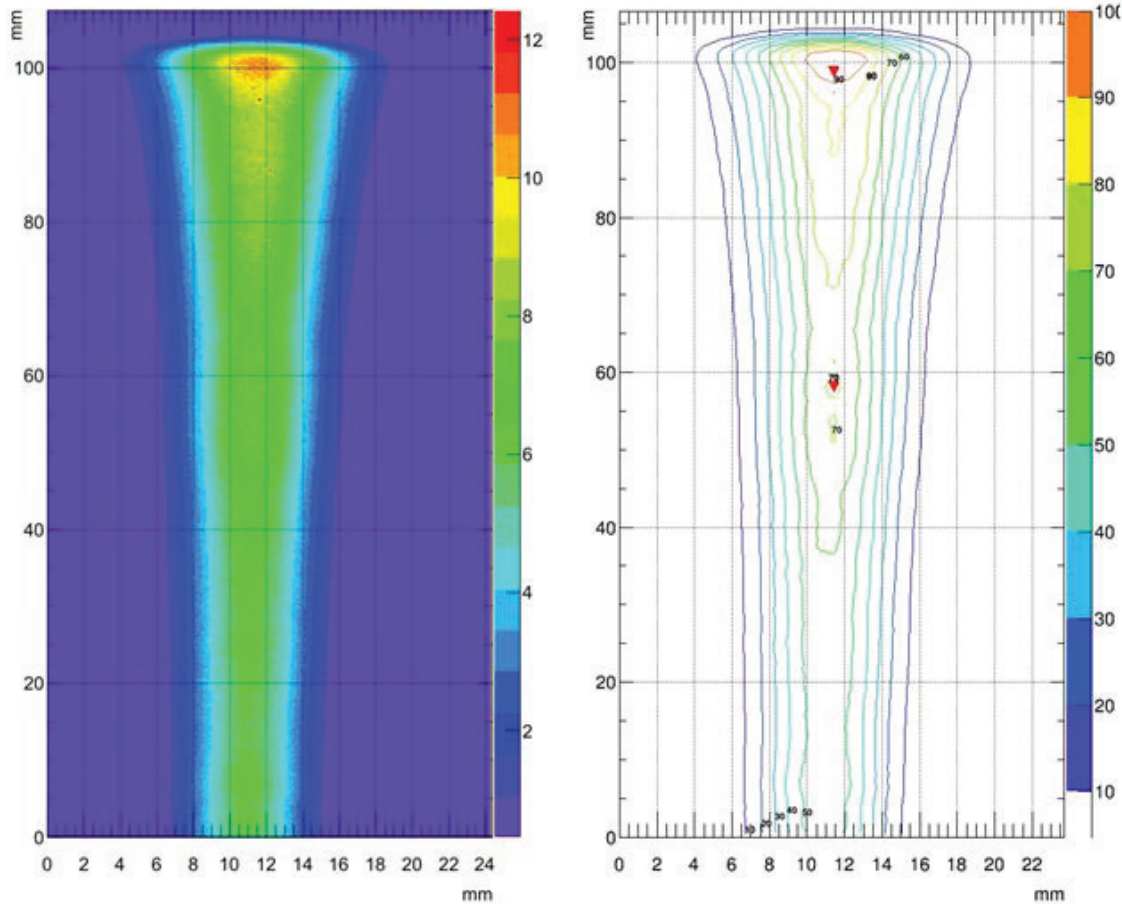
INTRODUCTION

Introduction

Proton therapy (PT) is one of the most accurate and modern methods of radiotherapy and radiosurgery. [1-3]. Protons can reduce the radiation load on surrounding tissues up to 30-50% in comparison with gamma rays. The use of a proton beam for tumors located near critical organs, such as the brain stem, optic nerves, etc are particularly effective. Therefore, in cases of head and neck cancer, proton therapy is the most advantageous of the available types of treatment for many patients.

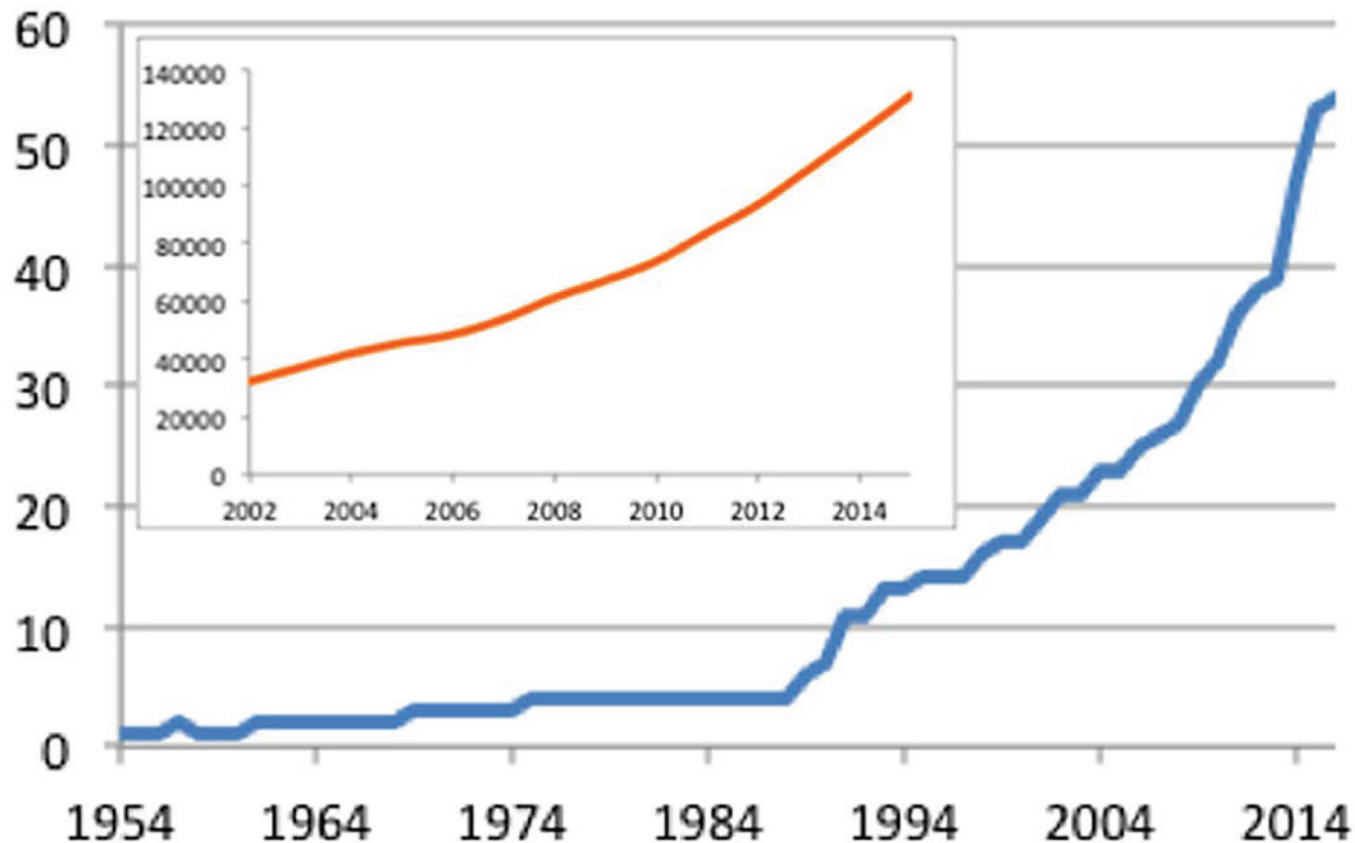


Introduction



Dose distribution for 120 MeV proton beam (with 10^9 protons/cycle intensity) in the water. From PTC Prometheus.

Introduction



Blue line – number of proton therapy centers in the world.

Orange line – number of treated patients in the world.

There is an increase of PT centers around the world [2].

Introduction

According to statistics, PT can be assigned to 50 thousand patients per year in Russia [3]. Unfortunately, in recent years, insufficient attention has been paid to this problem.

Proton Therapy Centers in Russia

Name and location of PT center	Start to treat
Medico-Technical Complex JINR, Dubna	1968
Protvino City Hospital, Protvino	2015
A. Tsyb Medical Radiological Research Centre (MRRC), Obninsk	2016
MIBS, Saint-Petersburg	2018

THE PROTON THERAPY COMPLEX «PROMETHEUS»

1.1. General Properties of the Complex

Table 1. Main characteristics of the accelerator complex

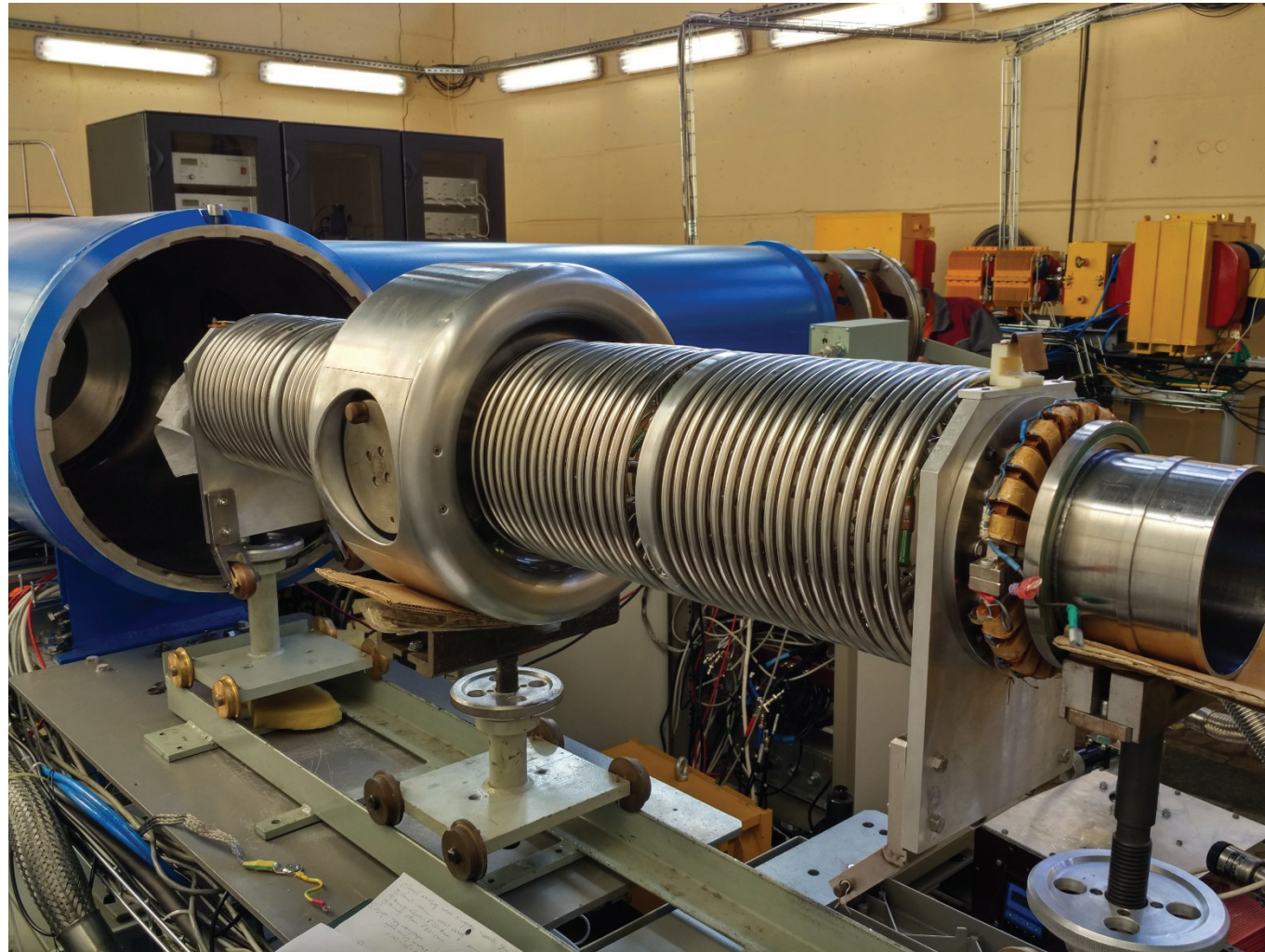
The range of accelerated proton energies, MeV	30 - 330
Magnetic field at injection, mT	80.66
Maximum magnetic field, T	1.9
Outer diameter of the ring, m	5
Accelerator weight, tons	15

1.2. The Injection Channel

The injector is designed for the initial production of protons and their acceleration to the energy of approximately 1 MeV.

The injector consists of a pulsed arc source of ions with a pulsed hydrogen inlet, electrostatic lenses, a tandem high-voltage accelerator and a 630 kV voltage source. With the help of a pulse valve, hydrogen enters the ion source, an electric arc is ignited, which leads to the formation of plasma. The electrostatic lens draws negatively charged hydrogen ions and electrons from the plasma. Electrons are discarded by a separator. Hydride ions enter a tandem accelerator, where they are recharged on a carbon film. Then, protons accelerated to 1 MeV are injected into the synchrotron. The voltage for the tandem accelerator is produced by a cascade generator.

1.2. The Injection Channel



1.3. The Synchrotron



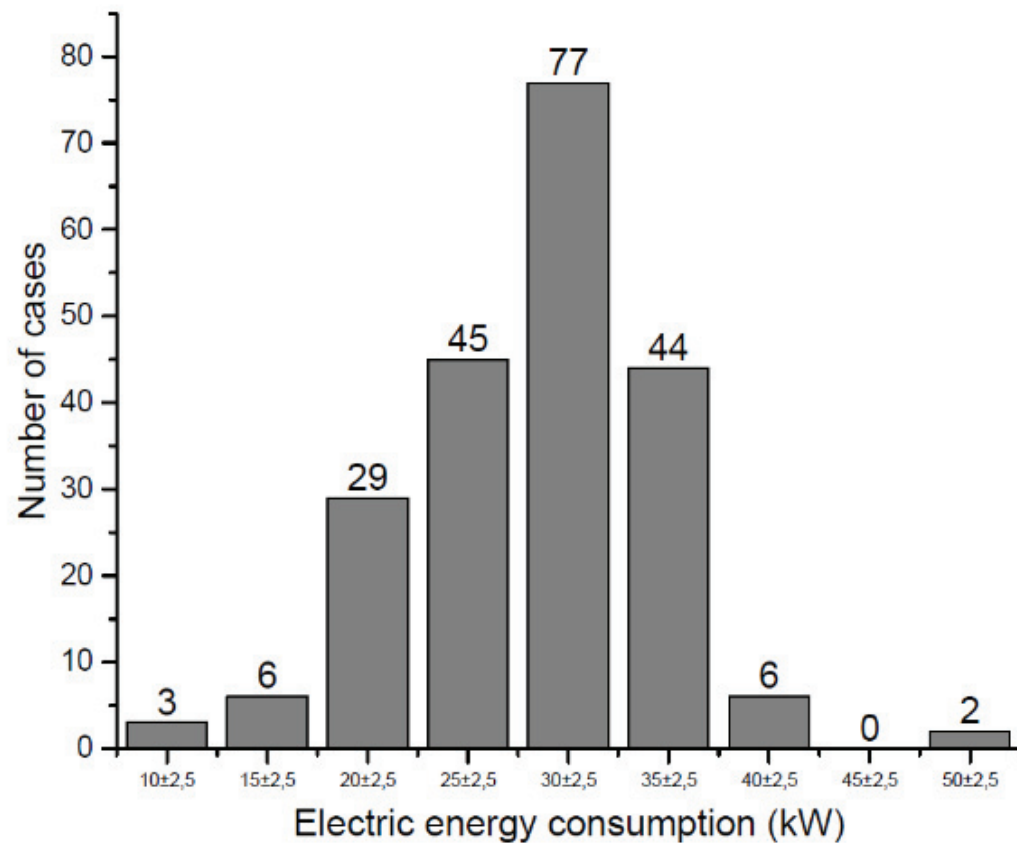
1.3. The Synchrotron

The synchrotron serves to accelerate the proton beam from the injection energy to the required energy in a given range. The synchrotron provides a high rate of particle acceleration equal to 330 MeV per 1.2 s.

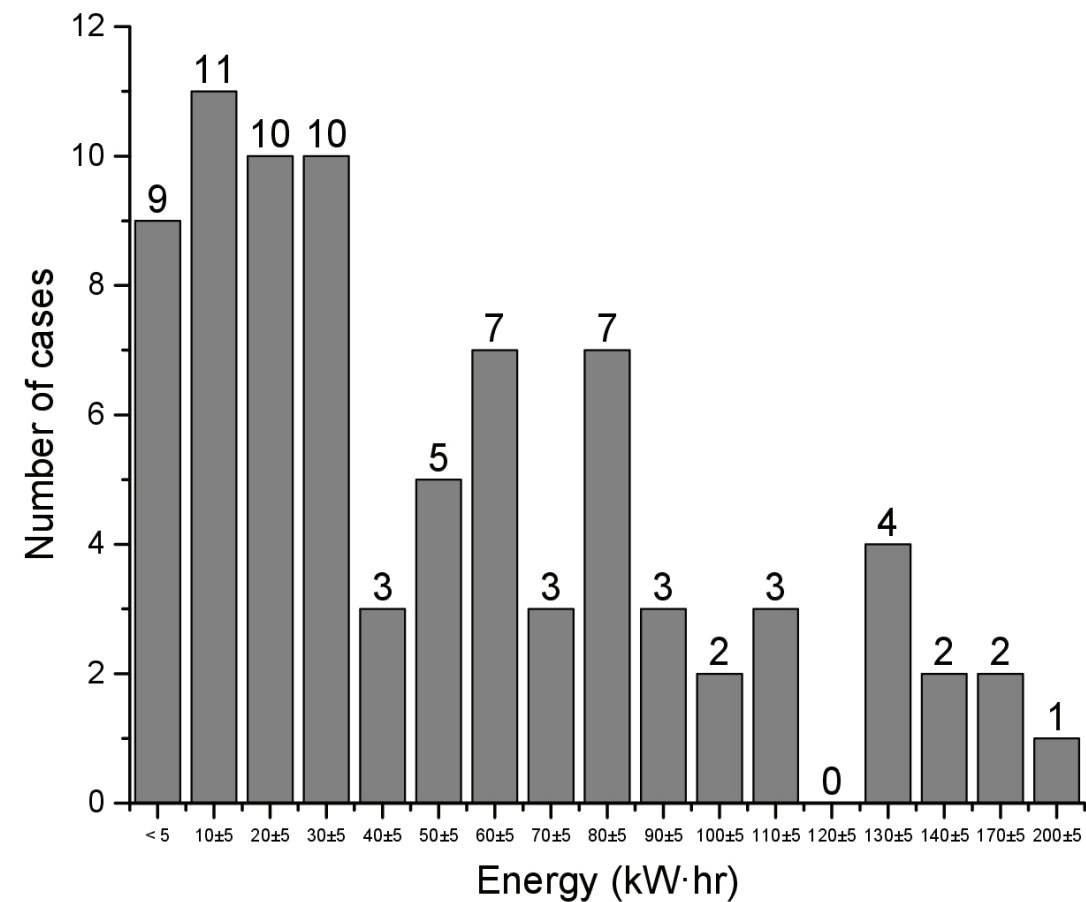
The magnetic system of the synchrotron is formed by four identical quadrants, separated by large free gaps. Each quadrant is formed by four iron blocks with parallel poles. Four magnets with a homogeneous field of each quadrant are arranged in a pairwise common winding.

During the calculation period, measurements of the power consumed by the magnetic system were made. The average value is 30 kW. The results integrated for each patient are shown in the diagram (next slide). The average value of the energy consumed for the full course of treatment of one patient is 56.8 kW·hr.

1.3. The Synchrotron



The diagram of the average electric energy consumption for each patient (total number is 211).



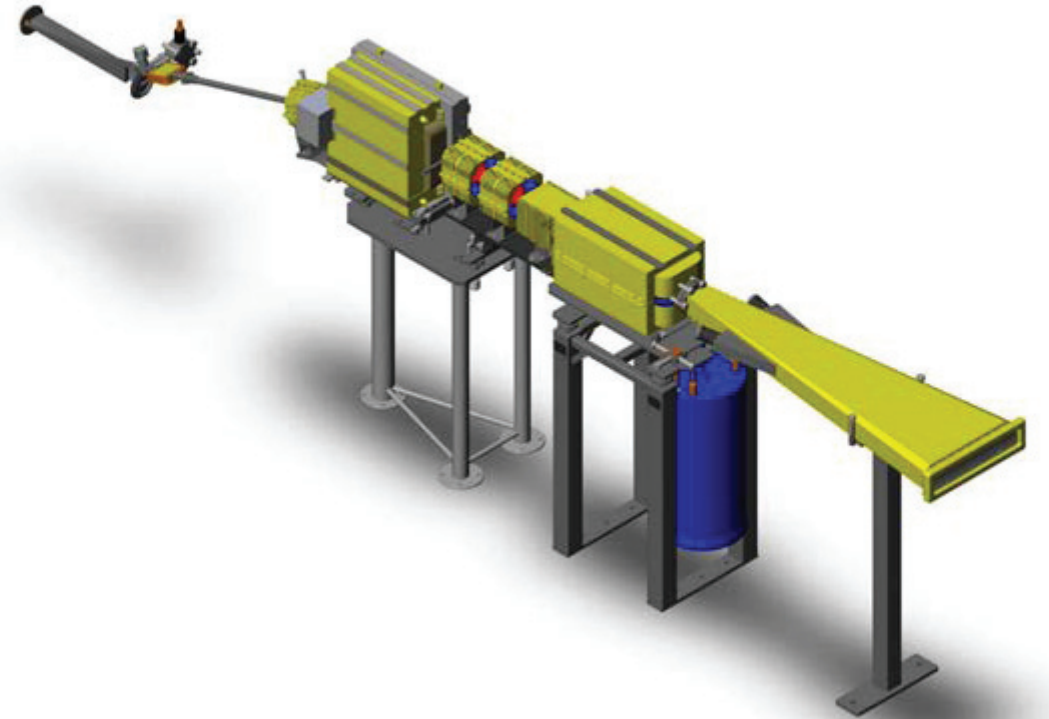
The diagram of the consumed energy for the complete treatment course of one patient (for 82 patients)

1.3. The Synchrotron

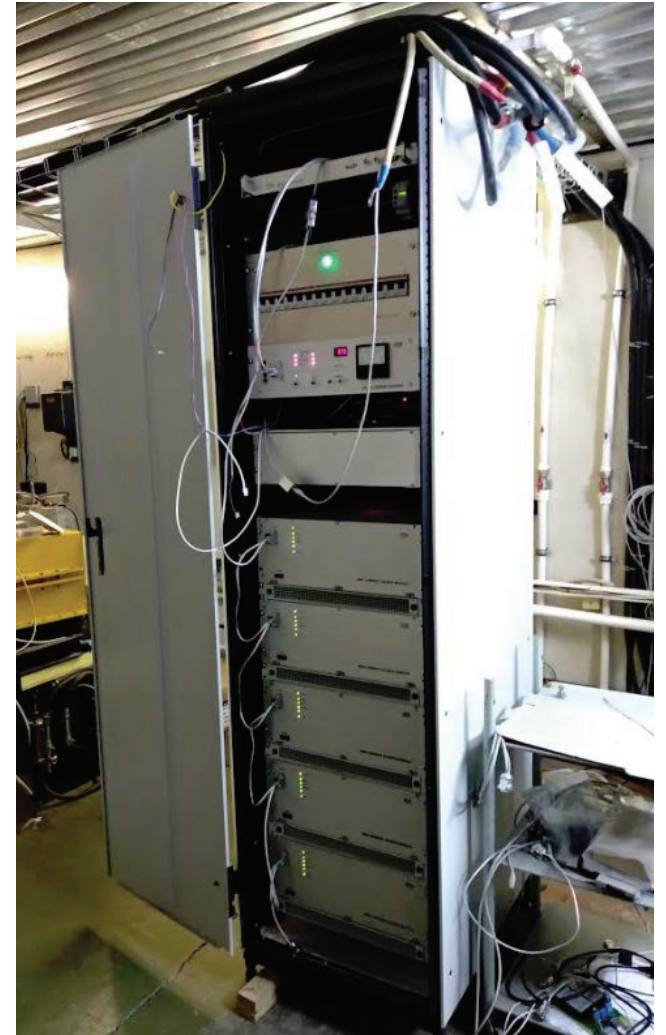


1.4. The Extraction Channel

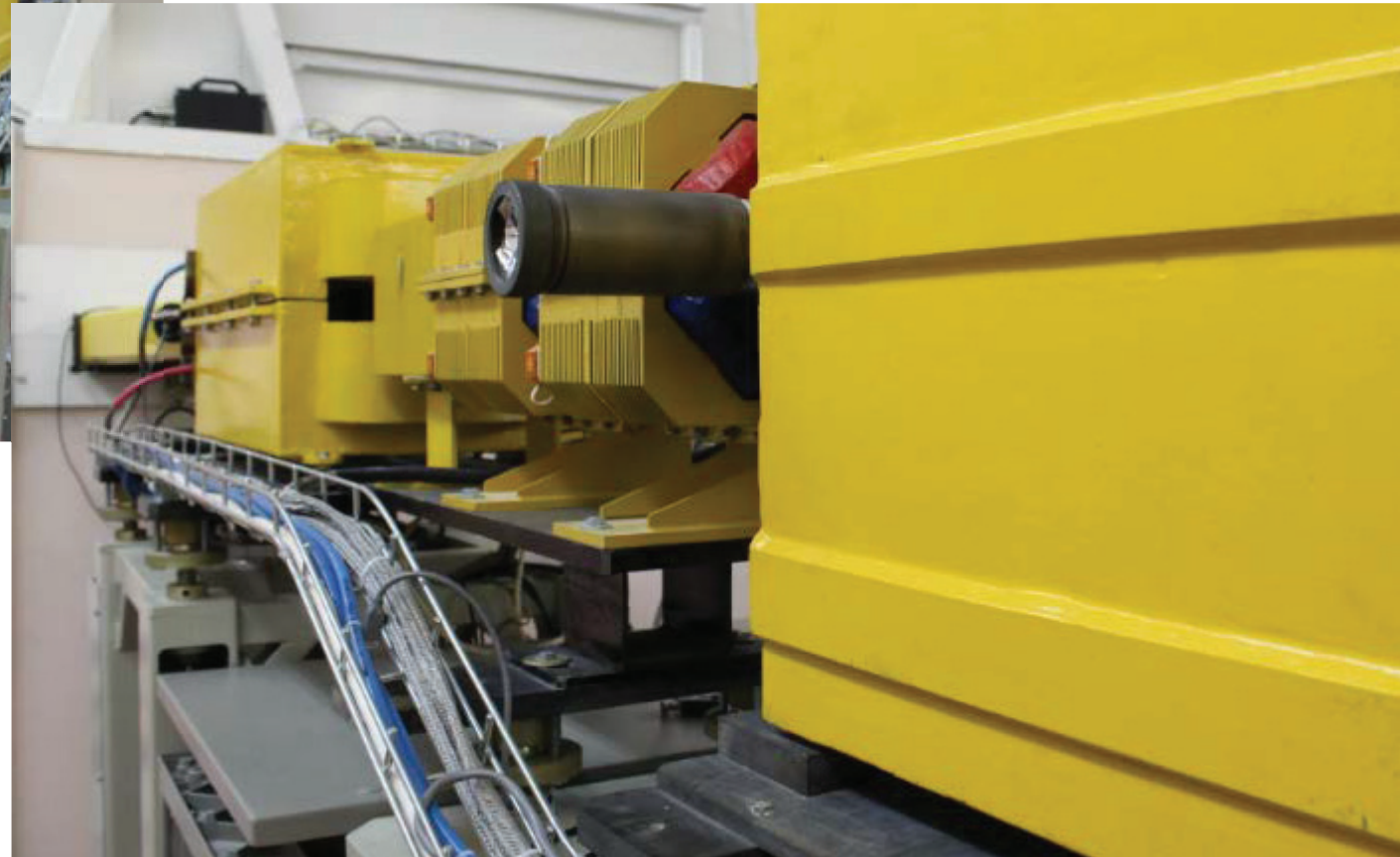
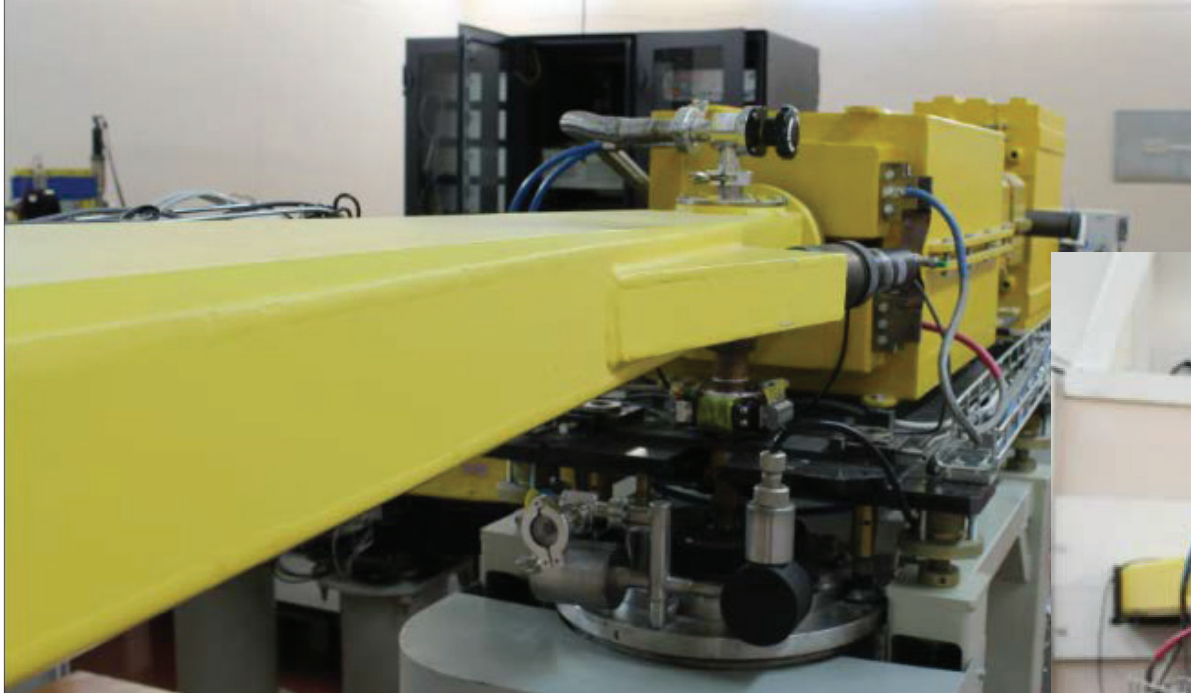
The beam extraction from the synchrotron occurs according to the following scheme: the buildup of betatron oscillations is initiated with subsequent scattering by the internal target, the beam enters the electrostatic deflector from the changed orbit, where it is radially thrown from into the Lamberson magnet. Observation of the beam is carried out from the phosphor-coated screens using the system of visual control. The area of active beam scanning on the target is $90 \text{ mm} \times 700 \text{ mm}$.



1.3. The Synchrotron Power Supply System



1.4. The Extraction Channel



1.4. The Extraction Channel

Formula (1) is used to describe the spatial distribution of proton beam. Parameters of extracted beam (the beam intensity was 10^9 protons per cycle) Gaussian function are shown in Table.

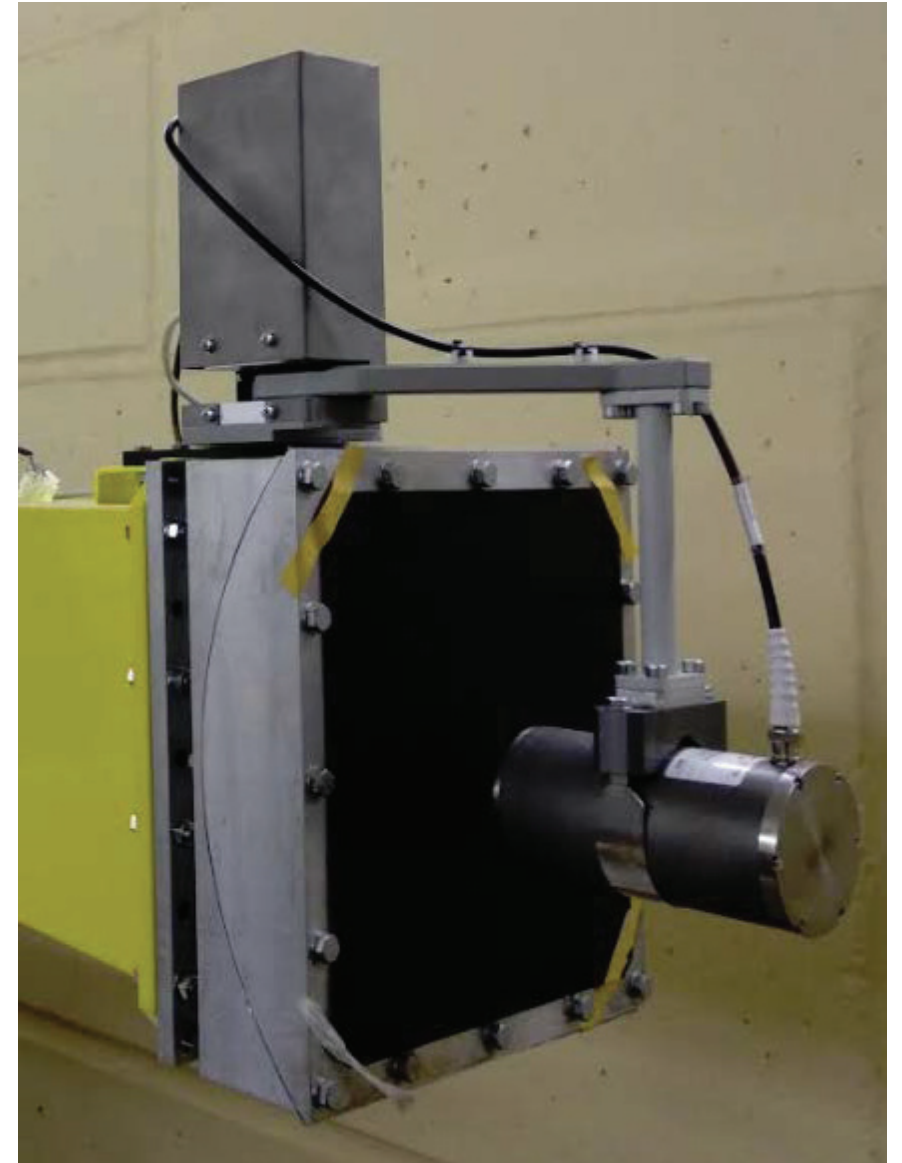
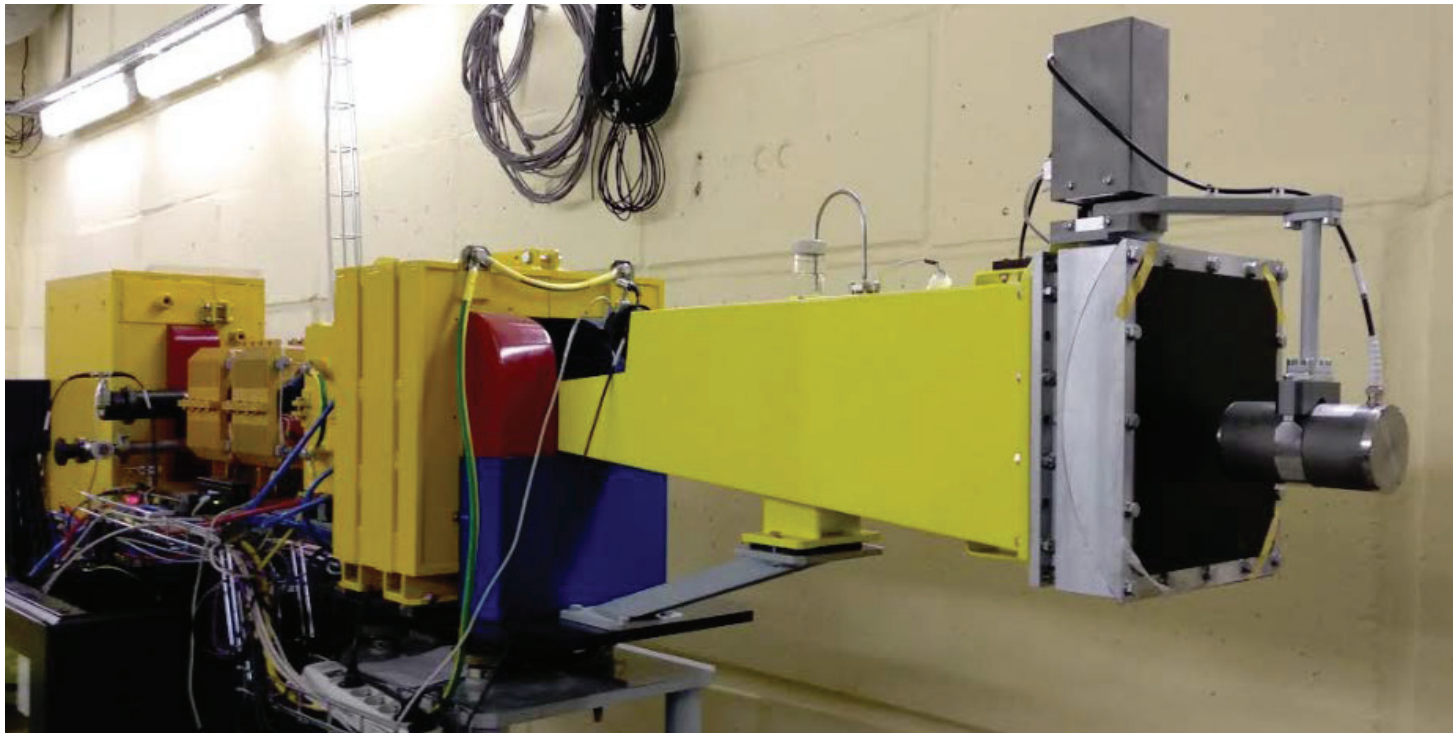
$$f(x) = A \cdot e^{-\frac{x^2}{2 \cdot \sigma^2}} \quad (1)$$

Parameters of extracted beam Gaussian function for different energies.

E, MeV	A_x	σ_x , mm	A_y	σ_y , mm
70	96	3.8	98	6.4
100	173	3.4	167	4.9
150	242	3.0	247	4.7
200	245	2.8	244	3.9
250	271	2.2	247	3.5

1.4. The Extraction Channel

Development of new pencil beam scanning system



1.5. The Patient Immobilization System

The system includes an armchair designed to fix the patient and move him to the irradiation zone, an x-ray unit represented by a small-dose X-ray tube and a digital X-ray panel (detector). With their help, the X-ray photographs are taken with a subsequent process of reconstructing them into a three-dimensional image for subsequent irradiation planning [5].

Dimensions (X, Y, Z), mm	1530×3000×1840
Max. power consumption	600 W
Angle of rotation	360°
Adjustment of patient's growth, mm	500
Control opportunities	direct (wired) remote (wireless - Bluetooth) remote (LAN)
Max. patient weight, kg	135

1.5. The Patient Immobilization System



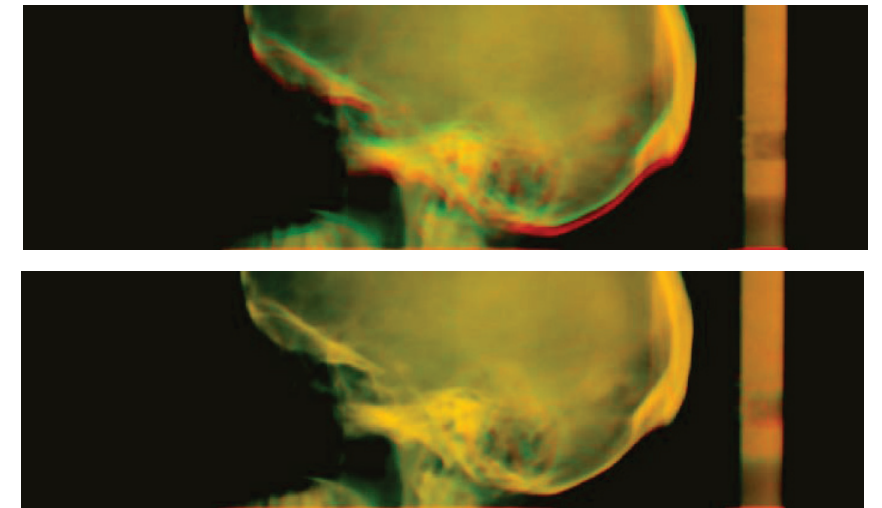
Patient Immobilization System in seated position.

1.5. The Patient Immobilization System



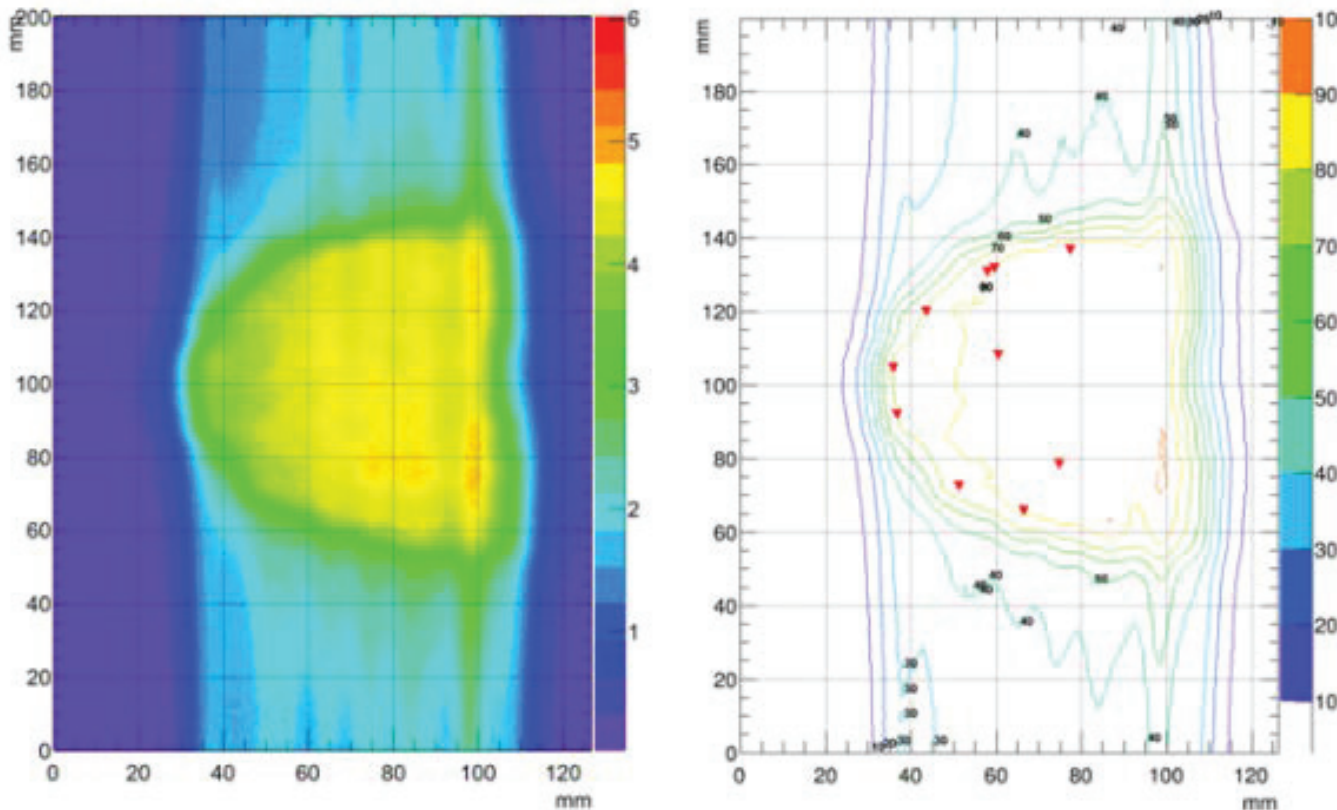
Patient Immobilization System in lying position.

1.5. The Patient Immobilization System



Patient position verification using X-ray CT

1.5. The Patient Immobilization System



Dose distribution for EBT3 film is irradiated in a solid-state phantom IBA in two intermittent directions in preparation for prostate irradiation. Preparation for patient irradiation in lying position.

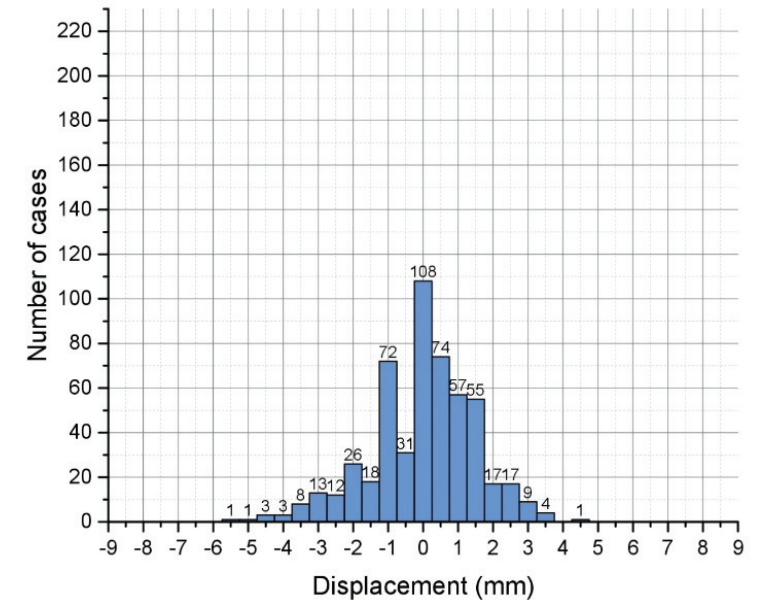
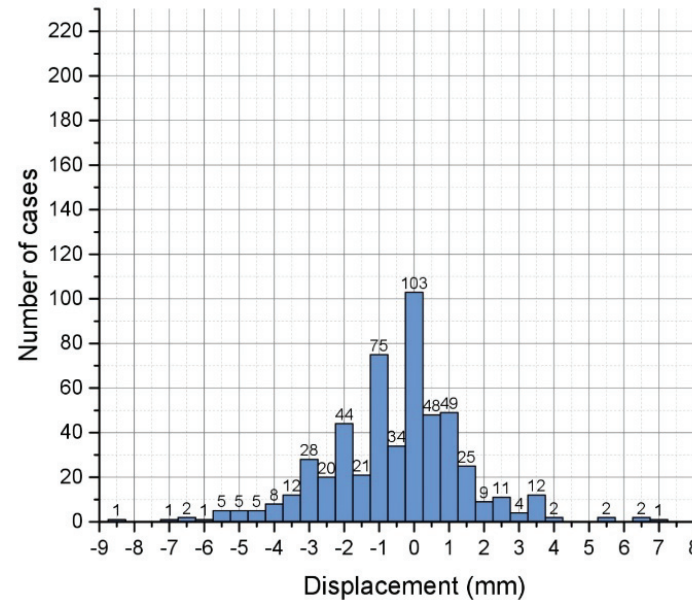
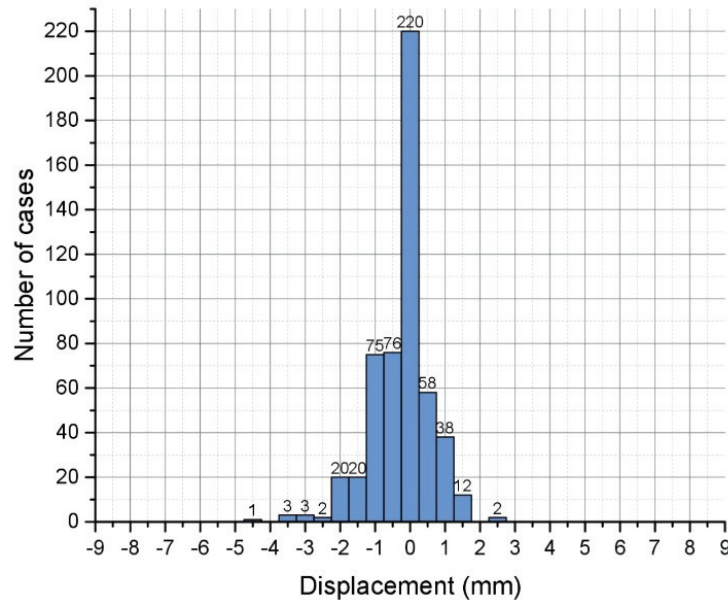
1.5. Radiographic system

The size of the X-ray detector, cm ²	40 × 40
The pixel size, mm ²	0.2 × 0.2
The system works in the modes	• "Snapshot" • "Radiography" • "Breath" • "Verification"



Radiographic system

1.5. The Patient Immobilization System



Patient setup displacement using patient immobilization system:

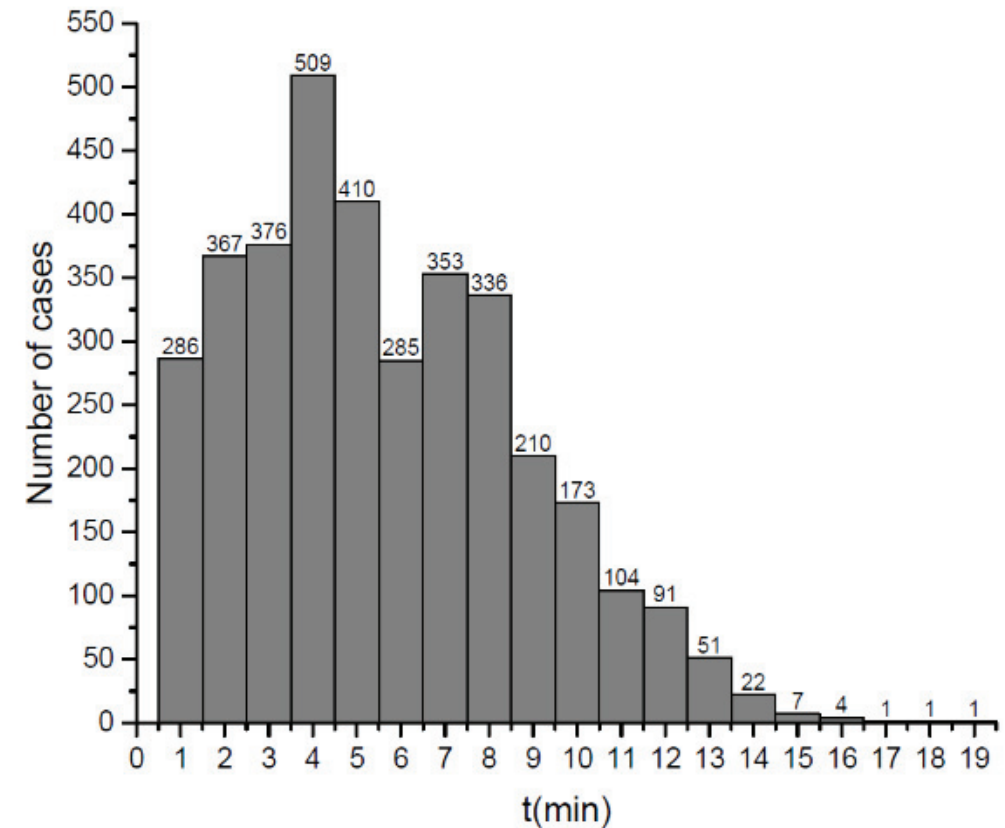
X-axis – right figure, Y-axis – middle figure, Z-axis – left figure

All displacement were eliminated after patient position verification procedure

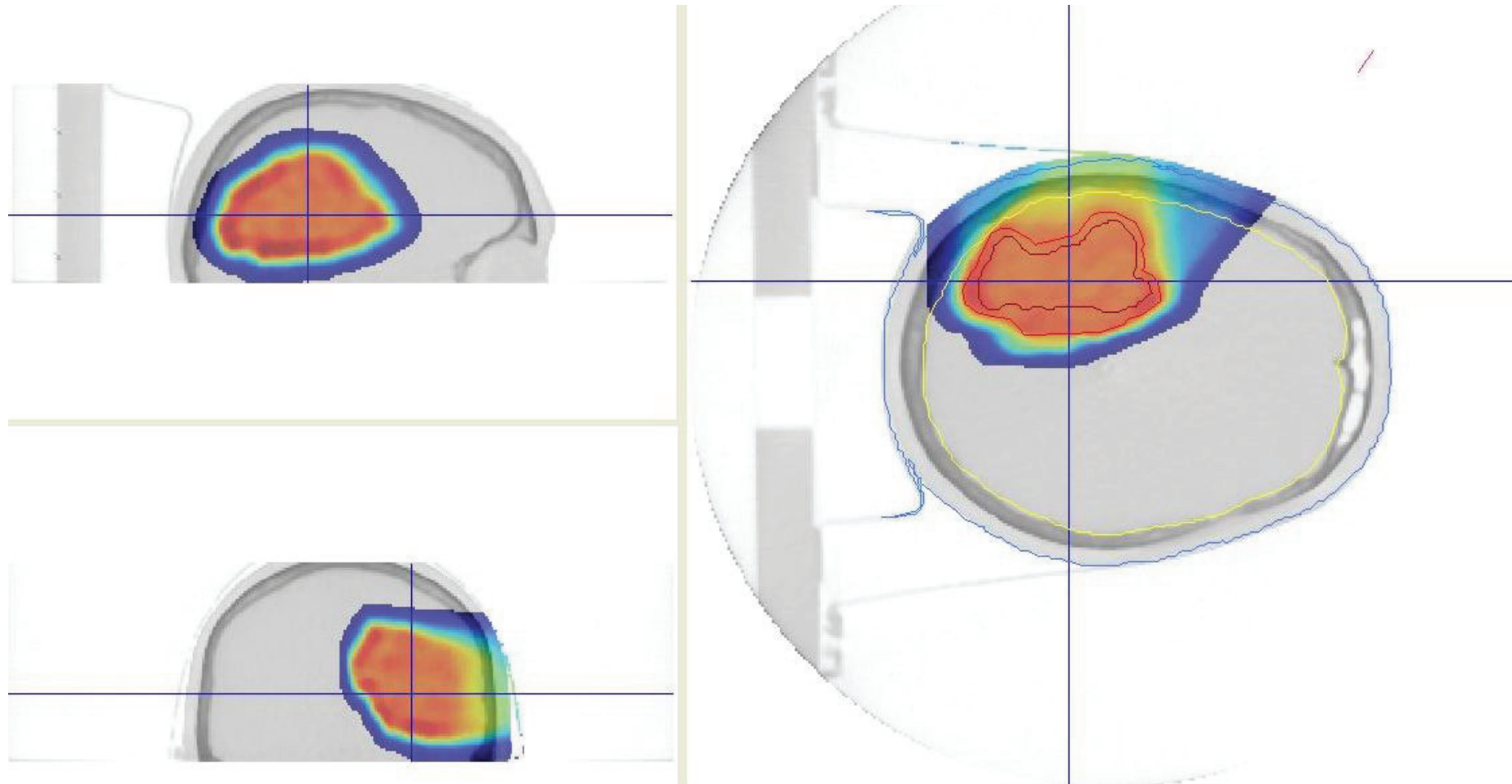
SOME RESULTS OF THE CLINICAL USE

2.1. The Irradiation Time of a Patient.

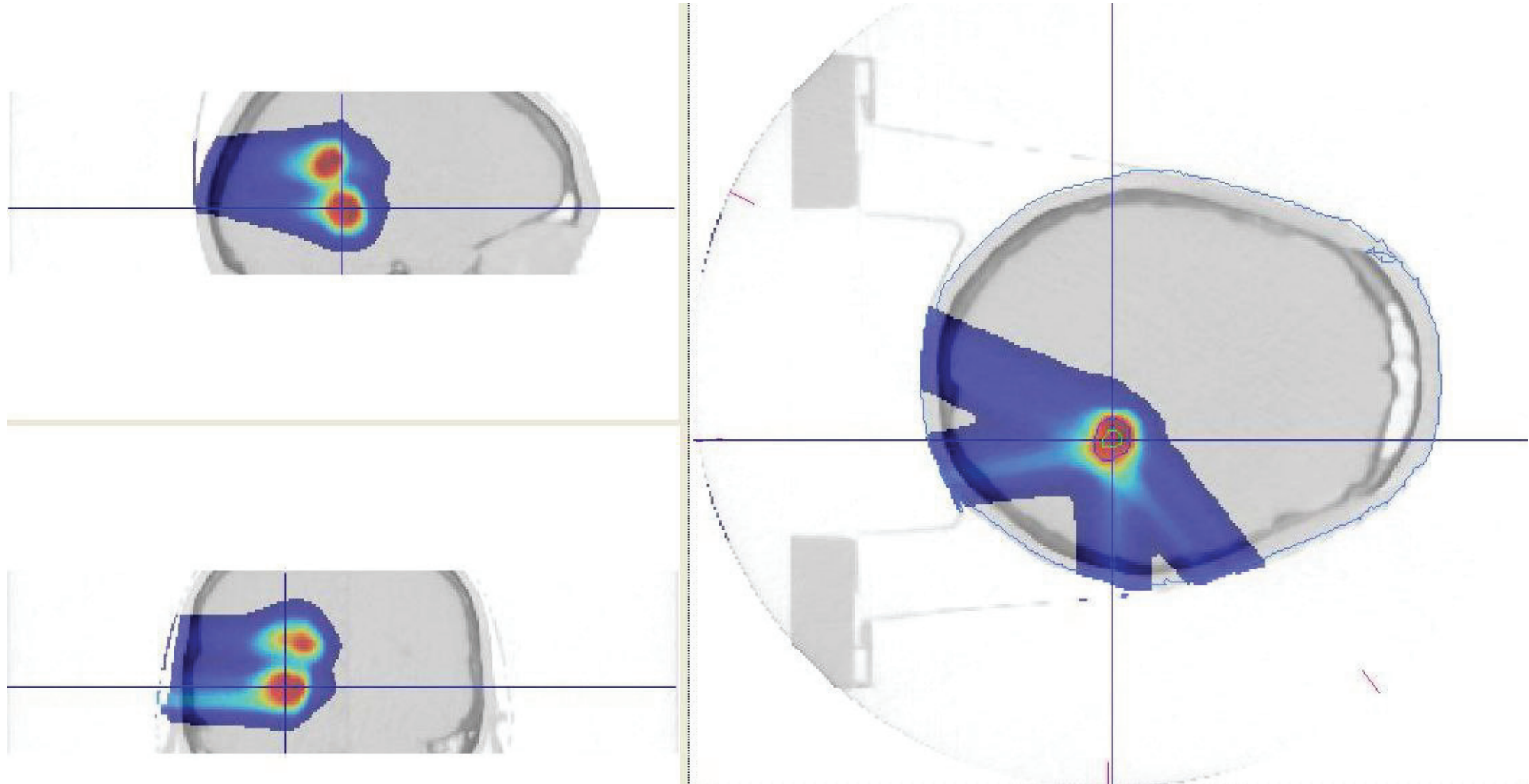
In real conditions, the irradiation time has not exceeded 15 minutes, and the average time was 5 minutes 52 seconds for 3587 cases, which easily meets the accepted standards. Average time of the patient stay in the treatment room was 10 minutes 12 seconds for first 82 patients [4].



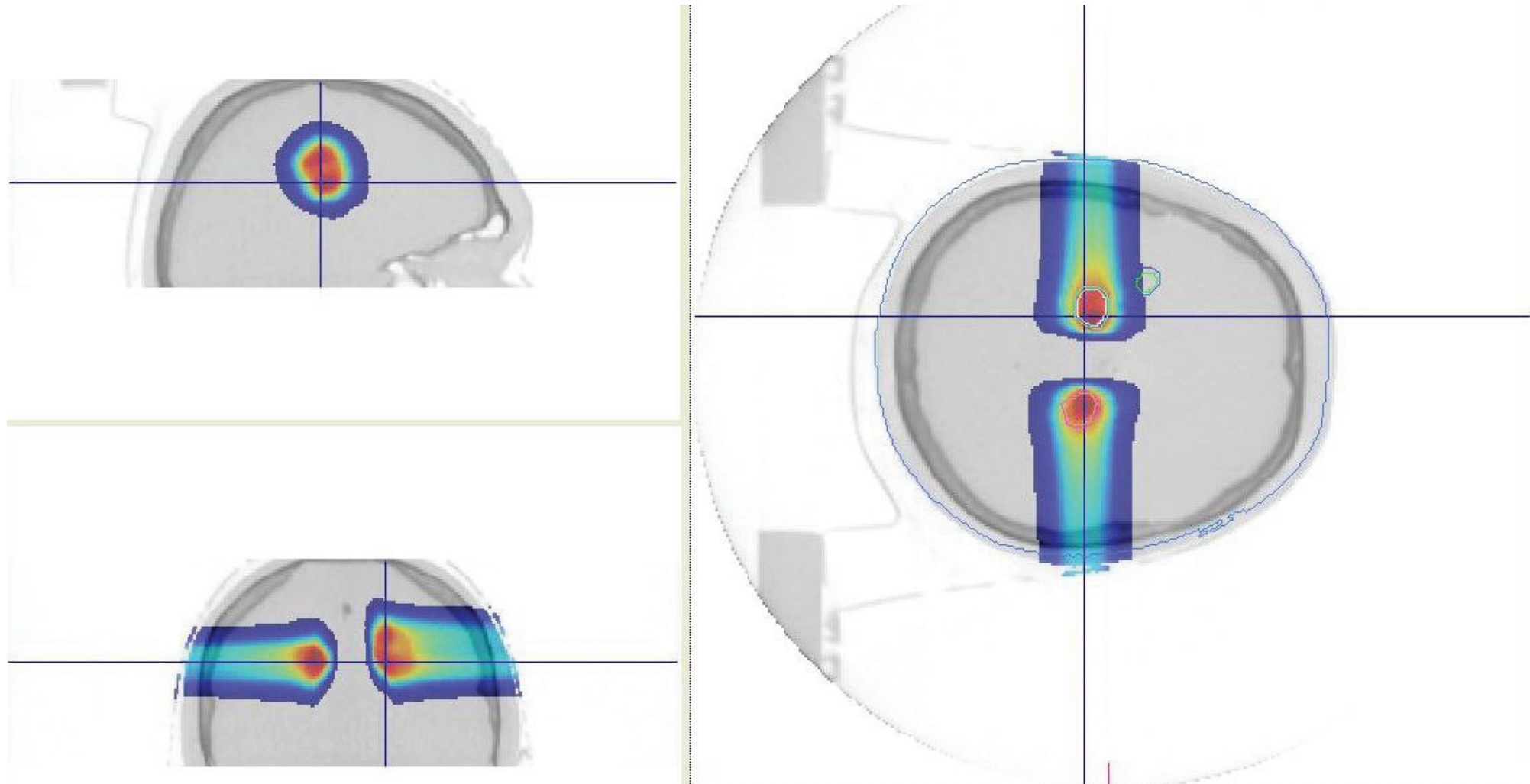
2.1. Real cases of head and neck cancer from clinical experience of using PTC Prometheus



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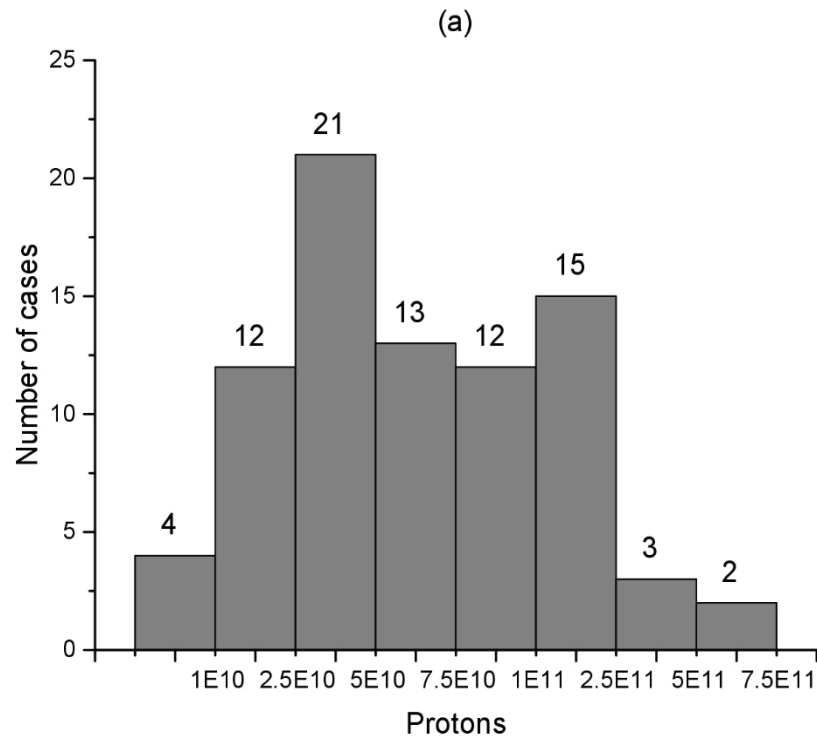


2.2. Number of particles per one irradiation session

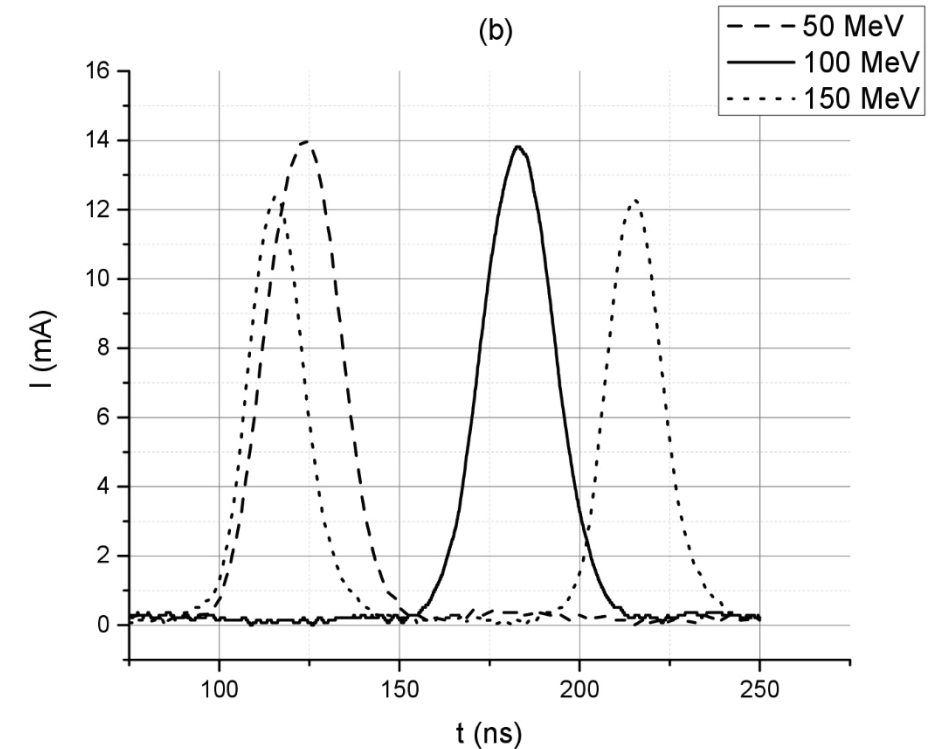
At first, it was necessary to prove that, in comparison with cyclotrons, the low intensity of the beam in the ring would be sufficient to conduct a radiotherapy session within the necessary time. Due to the high efficiency of the beam output from the accelerator, it was possible to achieve an average current value on the target for the entire energy range 0.5 - 1.5×10^9 prot./cycle. The standard cycle time is 2 s.

Average number of protons per one session is 9.1×10^{10}

2.2. Number of particles per one irradiation session



Number of particles necessary for one fraction.

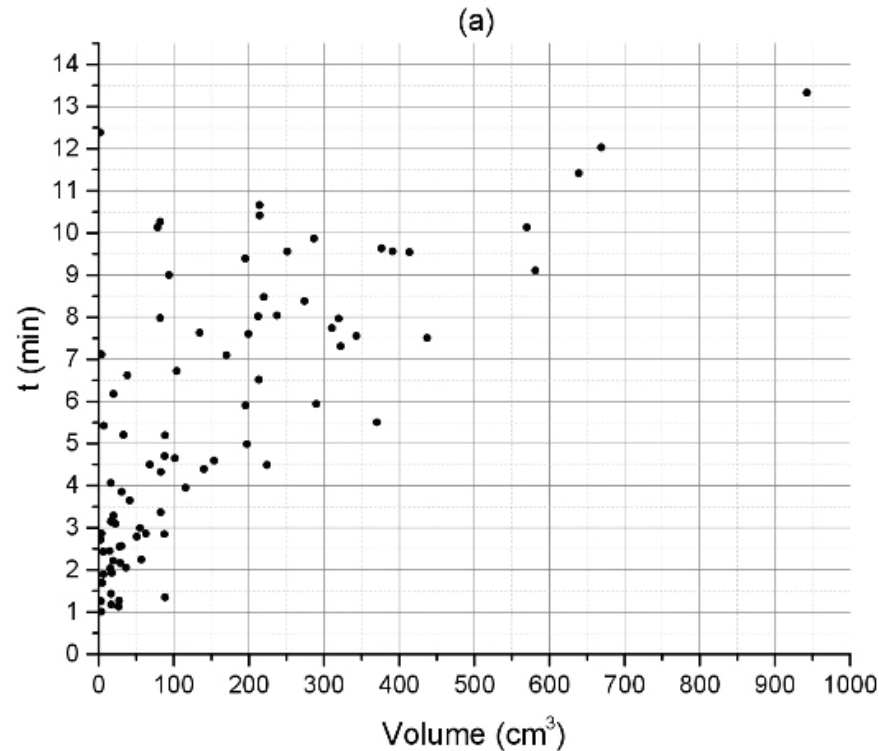


The beam current in the accelerator at the time of extraction.

2.3. Tumor volumes in case of head and neck cancer

It is believed that proton therapy is mainly used on neoplasms of small volumes (several dozens of cubic centimeters). Experience gained on PTC allows us to state the efficiency of the complex for the treatment of tumors of a much larger size. The average size of tumors was 145 cm³ for 212 cases.

2.3. Tumor volumes in case of head and neck cancer



Volume-time distribution (for 105 cases).

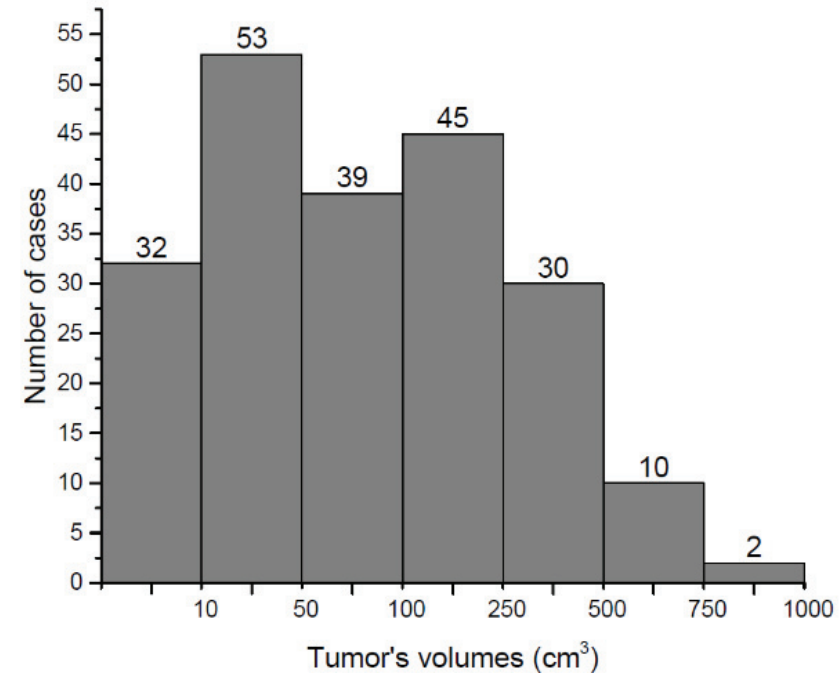
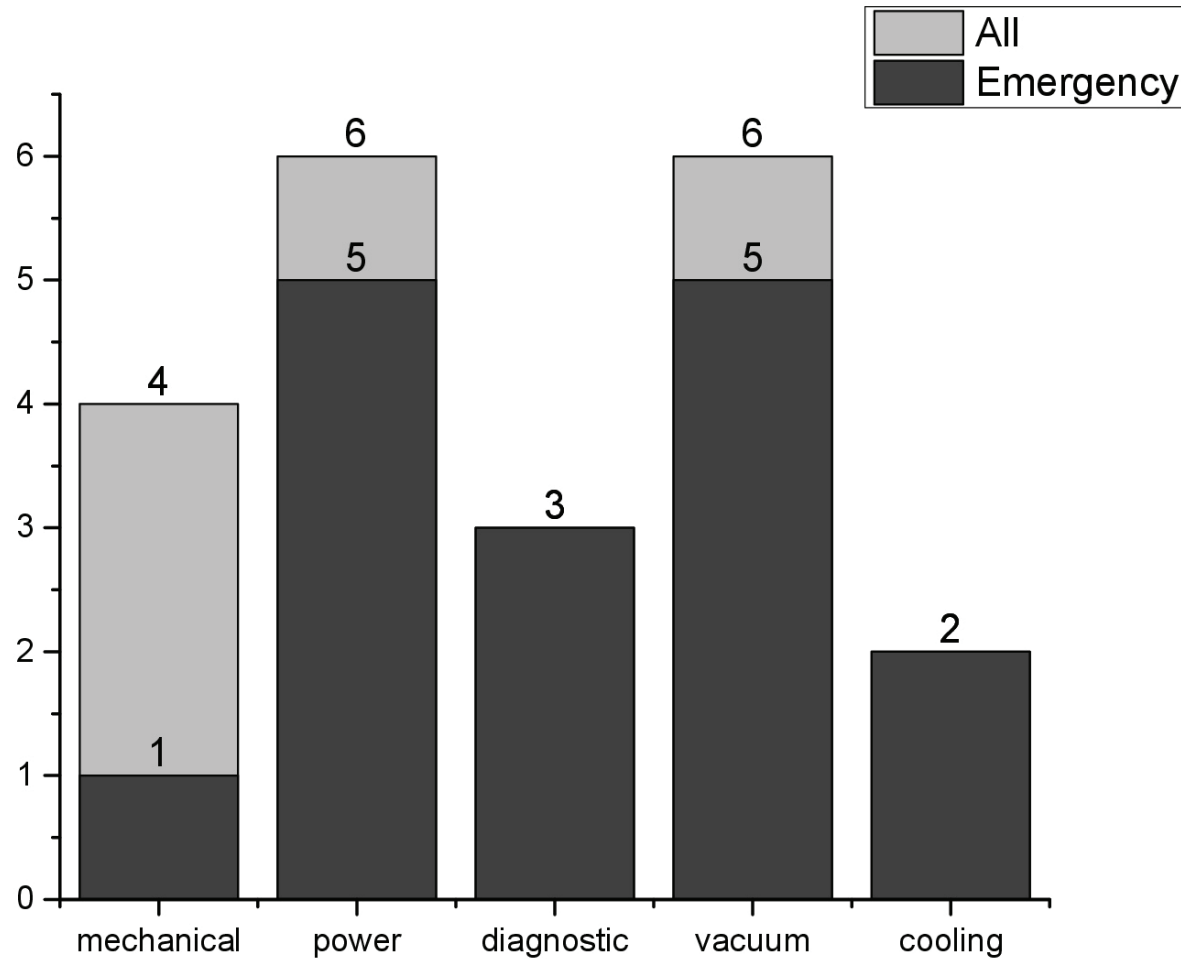


Diagram of the volumes of tumors that underwent the therapy (for 212 cases).

2.4. Repair schedule



For Protvino City Hospital,
Proton Therapy Center:

- 2 cases of cancellation of treatment
- 2 cases of delayed treatment (more than 2 hours)

ANALYSIS OF THE RESULTS OF TREATMENT

Analysis of the results of treatment

105 patients have a follow-up of sufficient duration (from 1 to 24 months) necessary for making preliminary conclusions about the results of the therapy. 19 of them have died during this period. There were three cases of development of radionecrosis. 7 patients have 24 months follow-up period. Acute radiation toxicity has not exceeded degree II (radiodermatitis III, mucositis I-II).

Analysis of the results of treatment

Treatment results for 86 patients.

Change in tumor volume after a course of PT			Number of cases
CR	Complete response	Decrease >70%	8%
PR	Partial response	Decrease 20-70%	32.5%
SD	Stable disease	Decrease <20%	39.5%
PD	Progression disease	Increase	20%

Analysis of the results of treatment

Types of tumors affected by PT (for 105 patients).

Types of tumors	Patient status after PT				
	CR	PR	SD	PD	D
Meningioma	1	1	13	2	1
Metastases		4	4	7	3
Glyoblastoma		2		2	7
Neurostezioblastoma	1				
Astrocytoma	1	6	5	4	4
Adenocarcinoma	1	1		1	3
Estesioneuroblastoma	1				
Hemangiopericytoma		2			
Skull Base Chordoma			3		
Pituitary adenoma		2	2		
Cancer of the salivary gland	1	1			
Flat cell carcinoma head and neck	1	5	1		
Ganglioglioma			1		
Schwannoma		1	3	1	
Melanoma			1		
Nasopharyngeal cancer		2			
Sarcoma		1			
Chondrosarcoma			1		
Mucosal cancer					1
Total:	7	28	34	17	19

Summary

Thus, PTC "Prometheus" has proved to be efficient and reliable in three years of clinical use in the treatment of head and neck cancer. If there is a developed infrastructure, the capacity of the facility can go up to 1000 people a year. The low weight, low power consumption and compact dimensions of the complex allow it to be placed in ordinary hospitals, without constructing separate buildings. In addition to that, PTC "Prometheus" was licensed to irradiate the entire human body in March 2017. Therefore, this complex is by far the only Russian development capable of solving the problem of proton therapy at the country level.

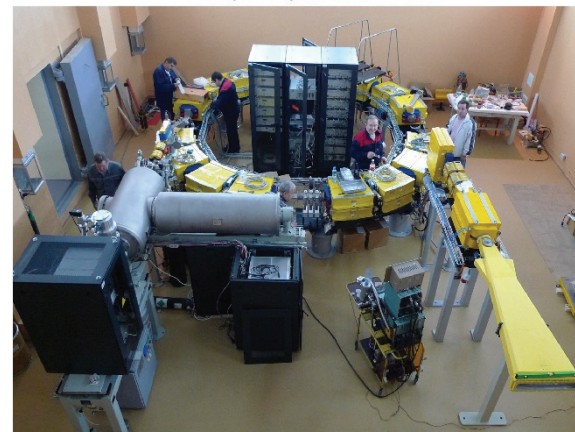


McLaren Hospital, Flint, USA

MGH, Boston, USA

МРНЦ им. Цыба г. Обнинск

Городская больница г. Протвино



Thank you for your attention

- [1] Paganetti H. Proton Therapy Physics, CRC Press, 2012, P. 651.
- [2] Paganetti H. Proton Beam Therapy, IOP Publishing, 2017, P. 23.
- [3] Particles and Nuclei, Letters. 2013. V.10, №7 (184). pp.1346-1375.
- [4] A.A. Pryanichnikov, V.V. Sokunov, A.E. Shemyakov, “Some results of the clinical use of the proton therapy complex “Prometheus””, In.: 12th International Scientific Workshop in Memory of Professor V.P. Sarantsev "Problems of Colliders and Charged Particle Accelerators", 2017.
- [5] V.E. Balakin et al., Clinical Application of New Immobilization System in Seated Position for Proton Therapy, KnE Energy & Physics | The 2nd International Symposium "Physics, Engineering and Technologies for Biomedicine", pp. 45–51, 2018